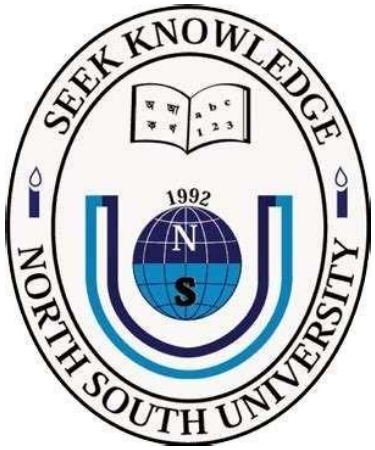
Analog Electronics II: Op-amp & 555 Timer



Course instructor:
Dr. Mohsin Sajjad
Assistant Professor
North South University

EEE 311

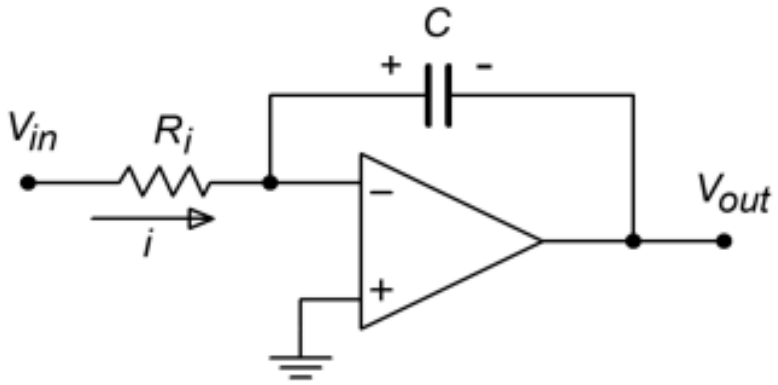
Lecture # 8

Textbook for Integrator and Differentiator

1. Fundamentals of Electric Circuits, Charles Alexander and Matthew Sadiku, 5th Edition: Chapter 6.6.1, 6.6.2 and relevant exercise problems.
2. Operational Amplifiers & Linear Integrated Circuits: Theory and Application, James M. Fiore 3rd Edition Chapter 10.1, 10.2, 10.3, and relevant exercise problems.

Integrator

An **integrator** is an op amp circuit whose output is proportional to the integral of the input signal.

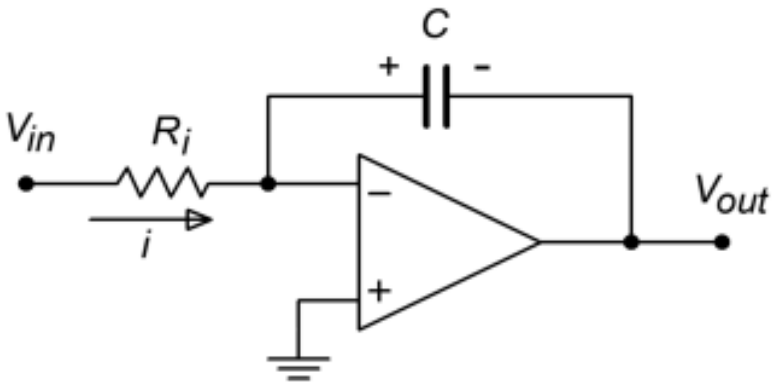


$$V_{out}(t) = -V_c(t)$$

$$i_R = \frac{v_i}{R}, \qquad i_C = -C\frac{dv_o}{dt}$$

$$i_R = i_C \Rightarrow \frac{v_i}{R} = -C\frac{dv_o}{dt}$$

$$dv_o = -\frac{1}{RC}v_i\,dt$$



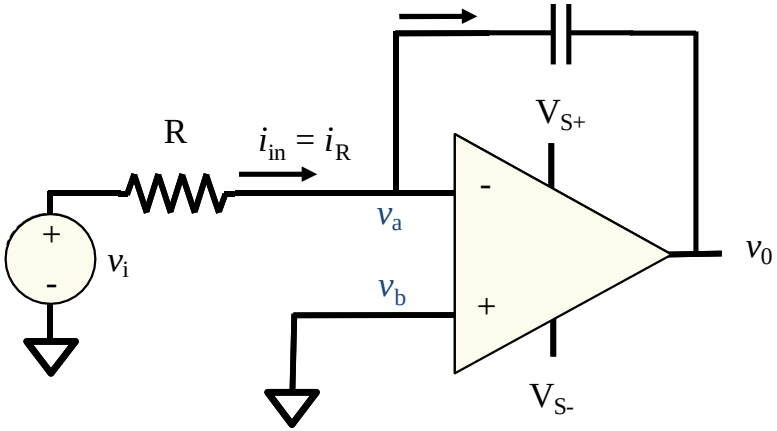
$$v_o(t) - v_o(0) = -\frac{1}{RC}\int_0^t v_i(\tau)\,d\tau$$

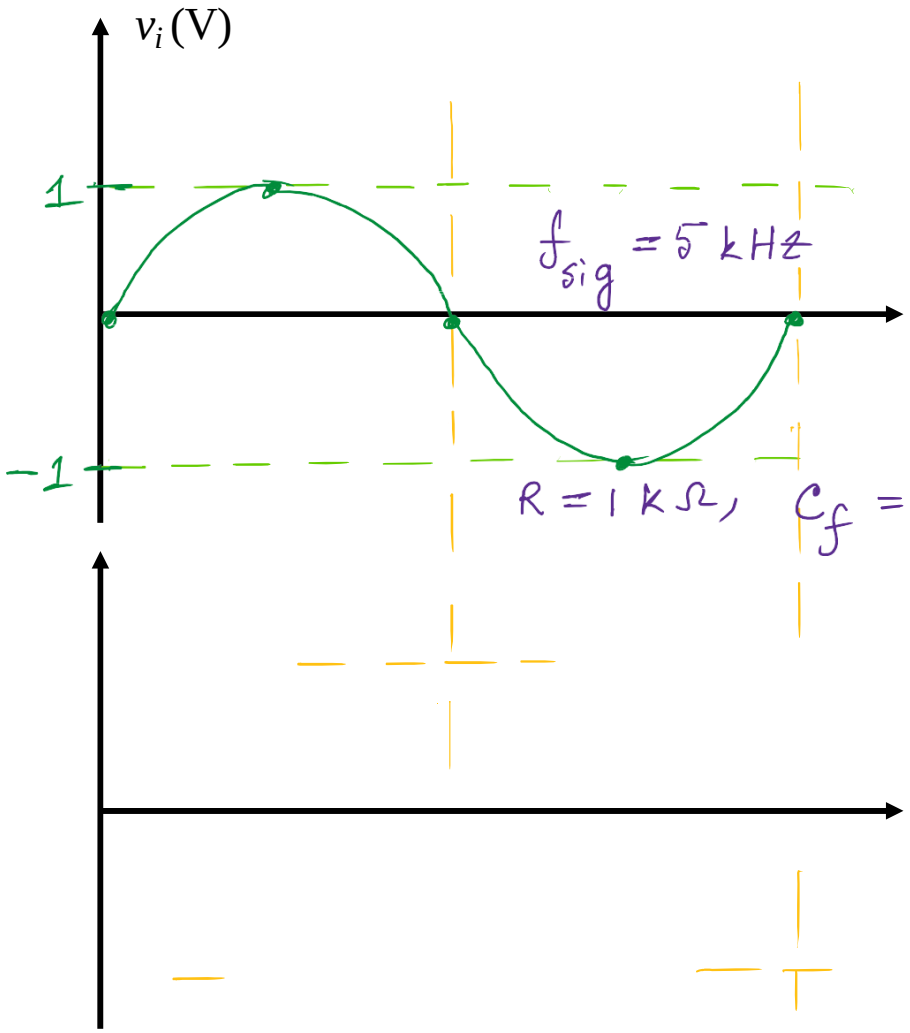
$$v_o = -\frac{1}{RC}\int_0^t v_i(\tau)\,d\tau$$

OR

$$V_{out}(t) = -\frac{1}{R_i C}\int V_{in}(t)\,dt$$

Basic Integrator





Example:

v_i is sin(xt) V.

v_0=?

Basic Integrator

Example:

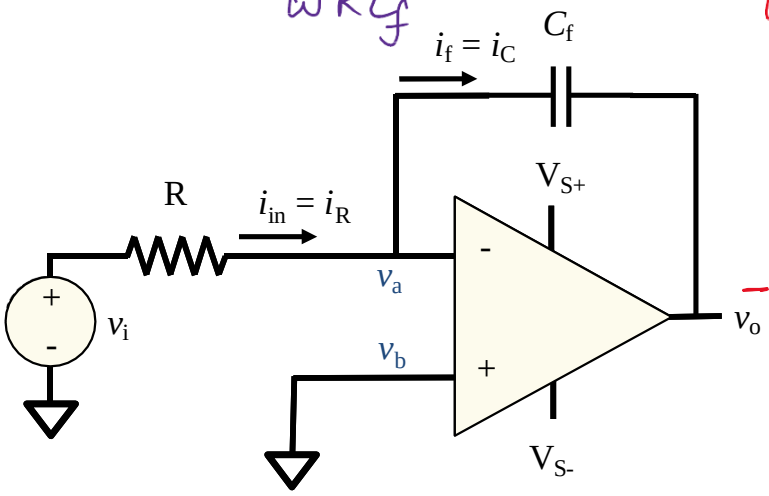
v_i is sin(xt) V.

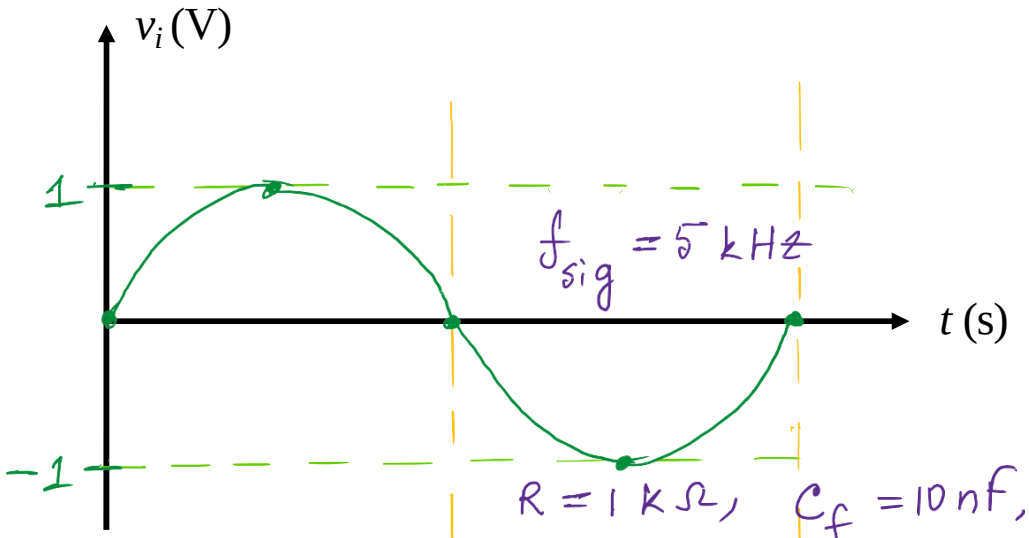
$v_o(0\,s) = 0\ V$

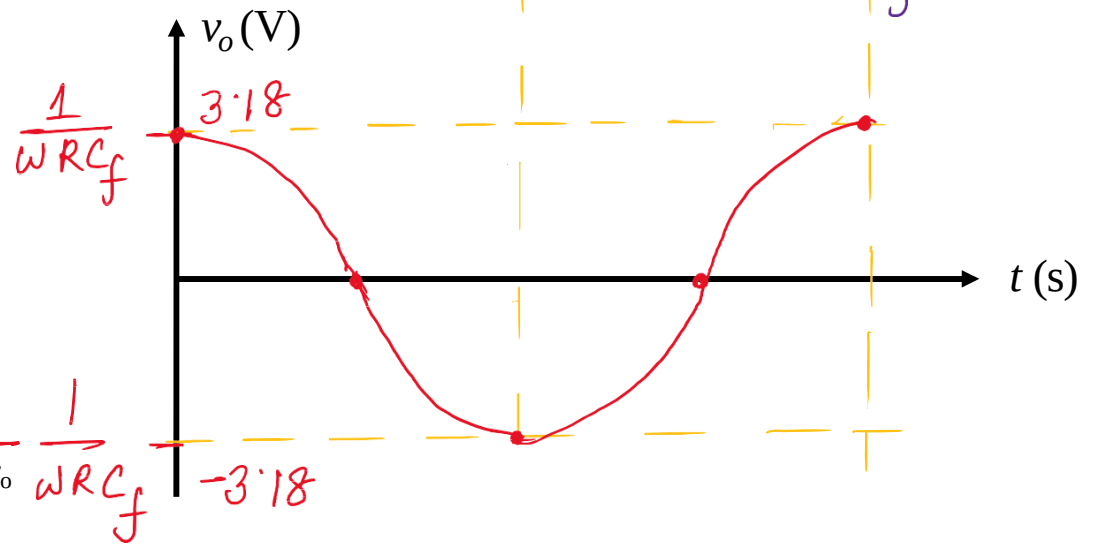
$$v_o(t) = -\frac{1}{RC_f}\int \sin(\omega t)\cdot dt$$

$$= -\frac{1}{RC_f}\left(-\frac{\cos\omega t}{\omega}\right)$$

$$= +\frac{1}{\omega RC_f}\cos\omega t$$







$$v_o = -\frac{1}{RC_f}\int v_i(t).dt$$

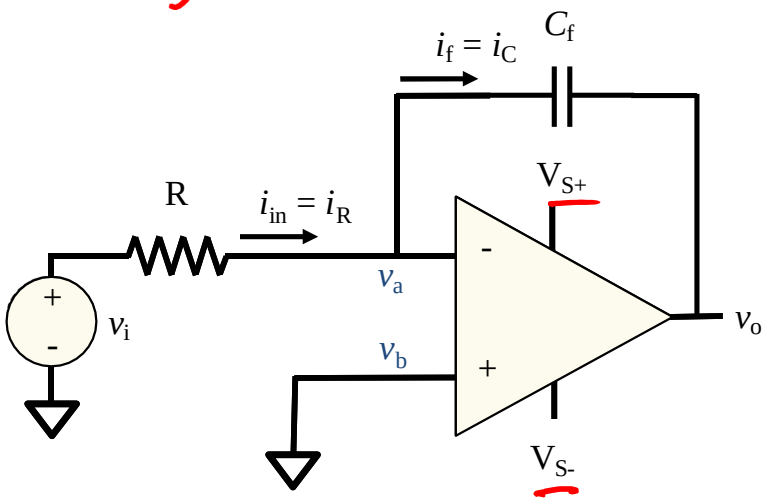
Basic Integrator

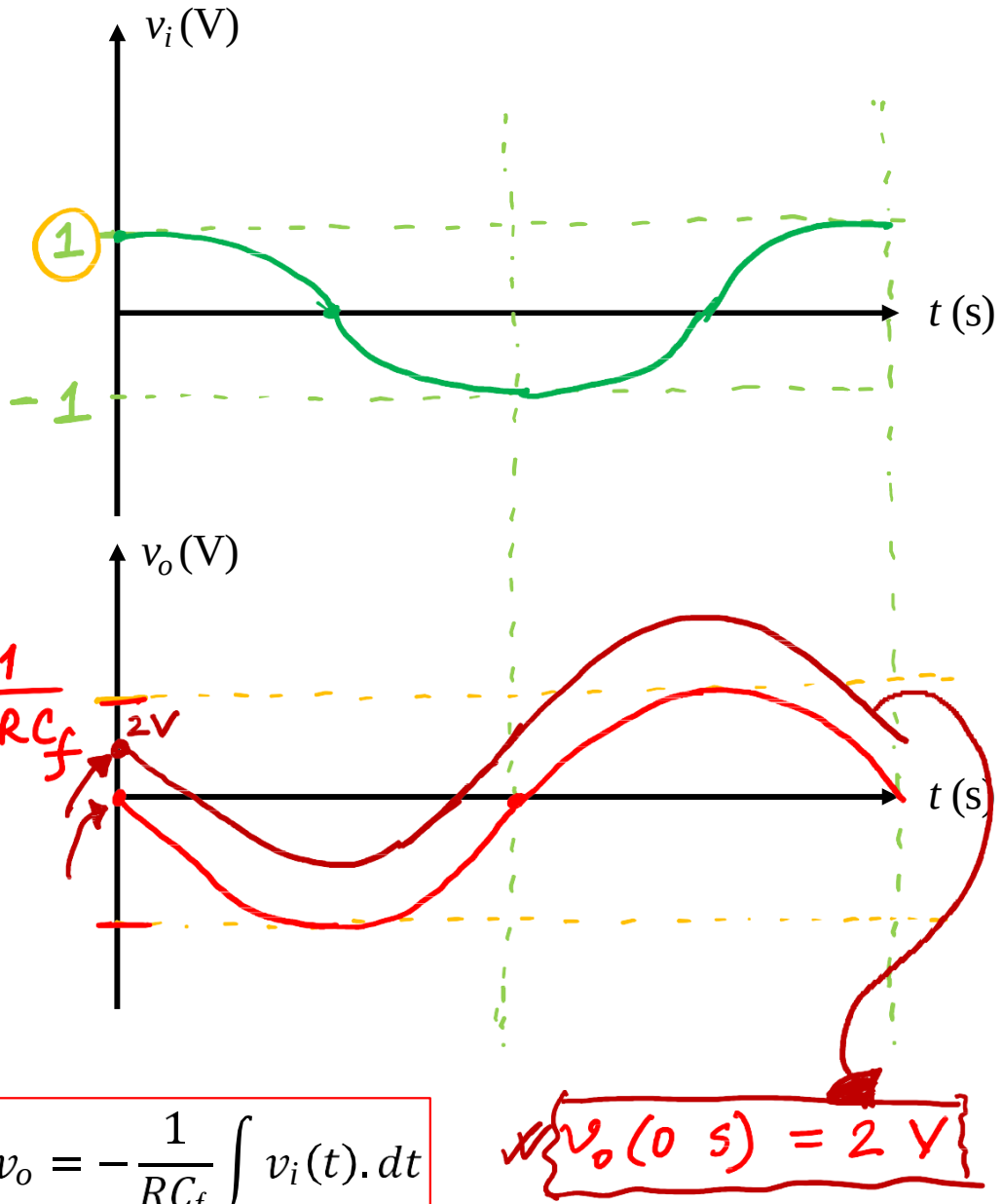
Example 3:

v_i is cos(xt) V.

$$v_o = -\frac{1}{RC_f}\int \cos(\omega t)\cdot dt$$

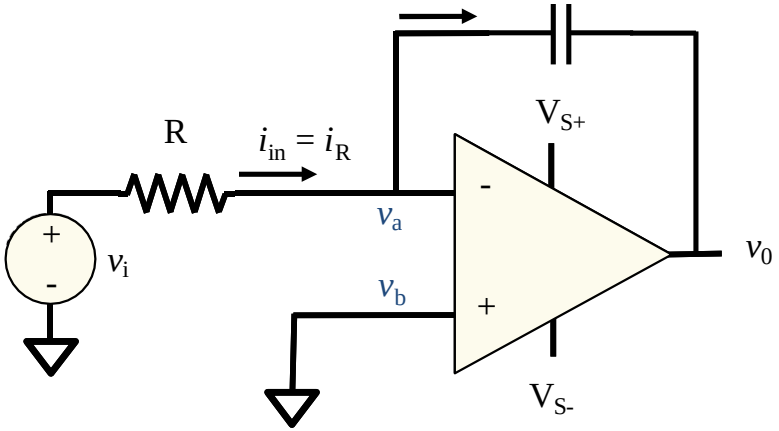
$$= \left(-\frac{1}{RC_f\omega}\right) \sin\omega t$$

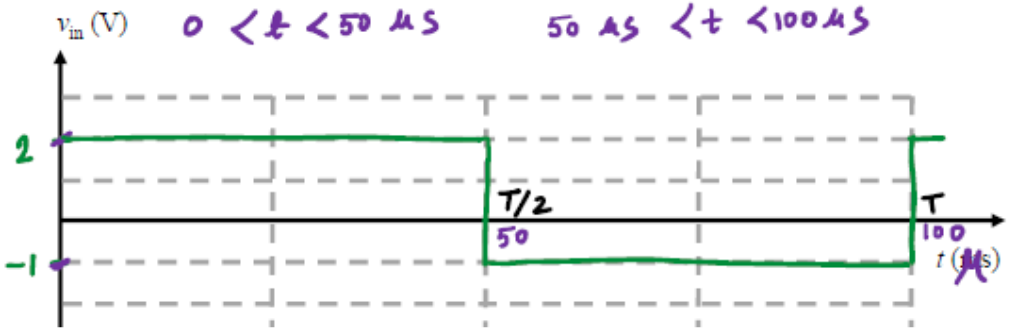




$$v_o = -\frac{1}{RC_f}\int v_i(t).dt$$

Basic Integrator





Example:

v_i is a square wave given in the figure.

v_0=?

Solution:

$C_f = 10\ nF$, $R = 1\ k\Omega$

$f = 10\ kHz$, $T = 100\ \mu s$

$v_0(t) = -\frac{1}{RC_f}\int 2 \cdot dt$, $0 < t < 50\ \mu s$

$v_0(t) = \underbrace{\left(-\frac{2}{RC_f}\right)}_{m_1} \cdot t$

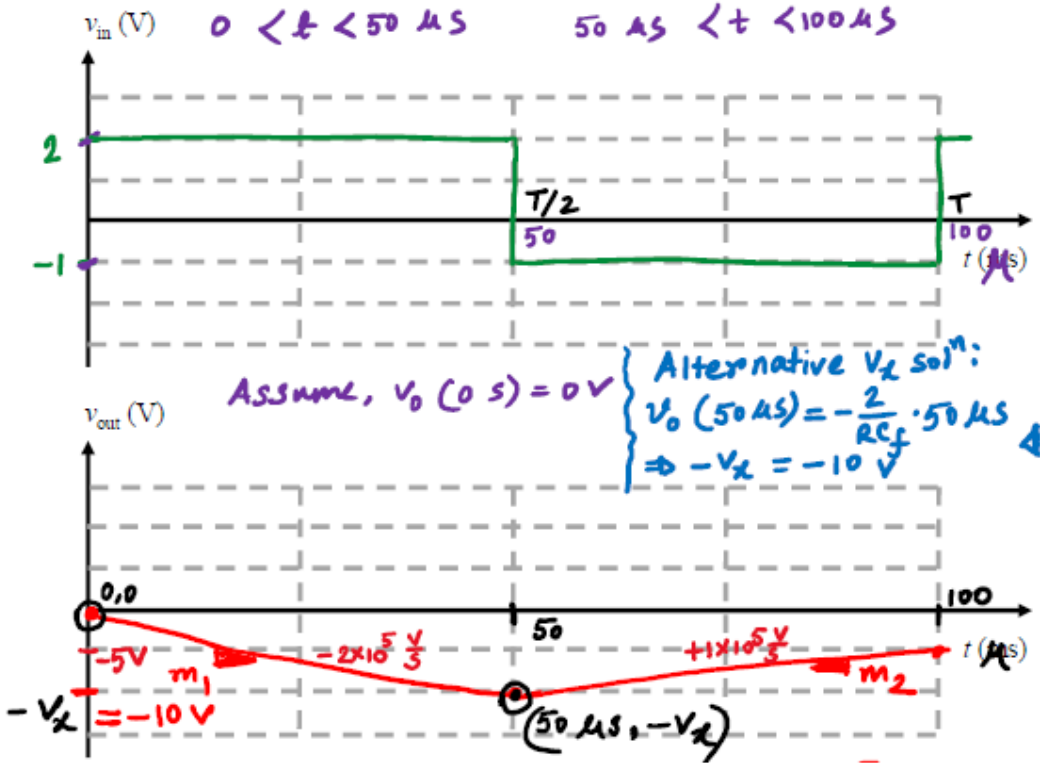
$v_0(t) = -\frac{1}{RC_f}\int (-1) \cdot dt$, $50\ \mu s < t < 100\ \mu s$

$v_0(t) = \left(+\frac{1}{RC_f}\right) t$

$m_1 = -\frac{2}{RC_f} = -\frac{2}{1\ k\Omega \times 10\ nF} = -2 \times 10^5$

$\frac{-V_x - 0V}{50\ \mu s - 0} = -\frac{V_x}{50 \times 10^{-6}}$

$m_2 = +\frac{1}{RC_f} = +1 \times 10^5$



$m_1 = -\frac{2}{RC_f} = -\frac{2}{1\,k\Omega \times 10\,nF} = -2\times10^5$

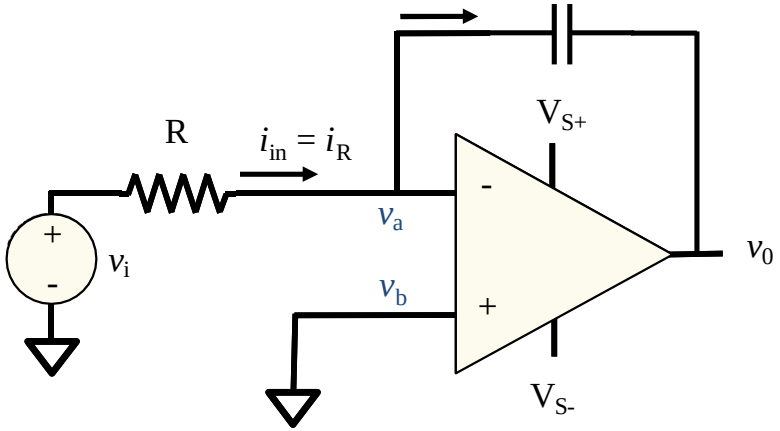
$\frac{-V_x - 0V}{50\,\mu s - 0} = -\frac{V_x}{50\times10^{-6}} = -2\times10^5$

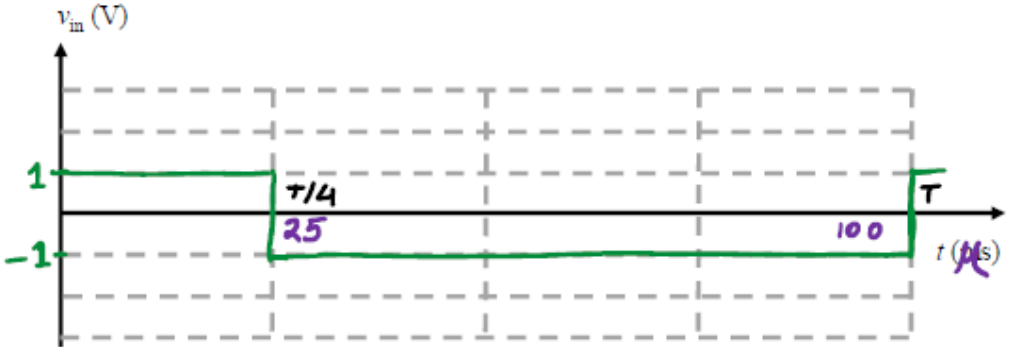
$V_x = 10V$

$m_2 = +\frac{1}{RC_f} = +1\times10^5$

$V_{x_2} = 5V$

Basic Integrator





Example:

v_i is a square wave given in the figure.

v_0=?

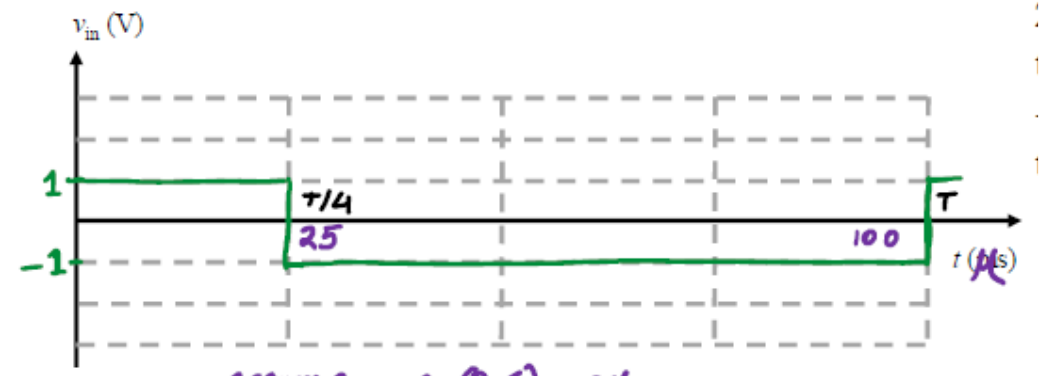
$v_0(t) = -\frac{1}{RC_f} \cdot t$, $0 < t < 25\,\mu s$

$v_0(t) = +\frac{1}{RC_f} \cdot t$, $25\,\mu s < t < 100\,\mu s$

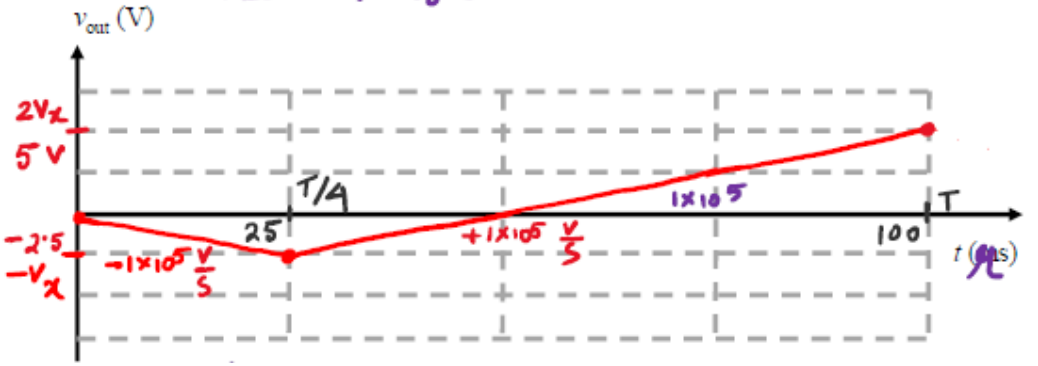
Calculation of $-V_x$:

$v_0(25\,\mu s) = -\frac{1}{RC_f} \cdot (25\,\mu s) = -\frac{1}{1\,k\Omega \times 10\,nF} \cdot 25\,\mu s$

$\Rightarrow$ $-V_x = -2.5\ V$ $\therefore$ $2V_x = +5\ V$



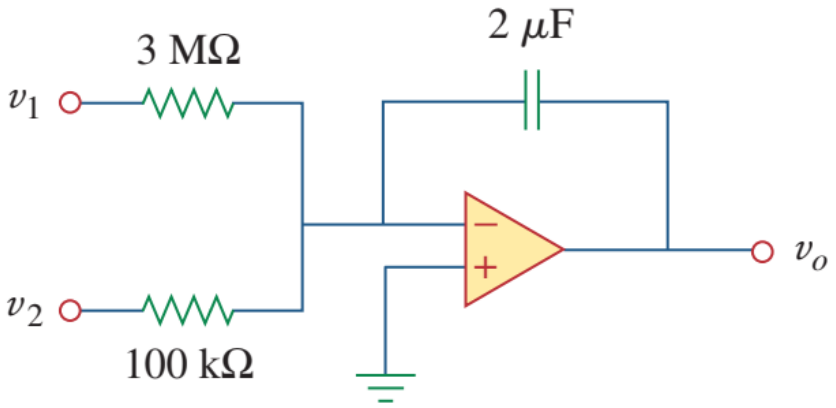
Assume, $v_o(0^-) = 0\,V$



Practice Problem

If $v_1 = 10 \cos 2t$ mV and $v_2 = 0.5t$ mV, find v_o in the op amp circuit

Assume that the voltage across the capacitor is initially zero.



Solution:

This is a summing integrator, and

$$v_o = -\frac{1}{R_1 C}\int v_1\,dt - \frac{1}{R_2 C}\int v_2\,dt$$

$$v_o = -\frac{1}{R_1 C}\int v_1\,dt - \frac{1}{R_2 C}\int v_2\,dt$$

$$= -\frac{1}{3\times 10^6 \times 2\times 10^{-6}}\int_0^t 10\cos(2\tau)d\tau$$

$$-\frac{1}{100\times 10^3 \times 2\times 10^{-6}}\int_0^t 0.5\tau d\tau$$

$$= -\frac{1}{6}\frac{10}{2}\sin 2t - \frac{1}{0.2}\frac{0.5t^2}{2} = -0.833\sin 2t - 1.25t^2\ \text{mV}$$

Practice Problem

Using a single op amp, a capacitor, and resistors of 100 kΩ or less, design a circuit to implement

$$v_0 = -50\int_0^t v_i(t)dt$$

Assume $v_o = 0$ at $t = 0$.

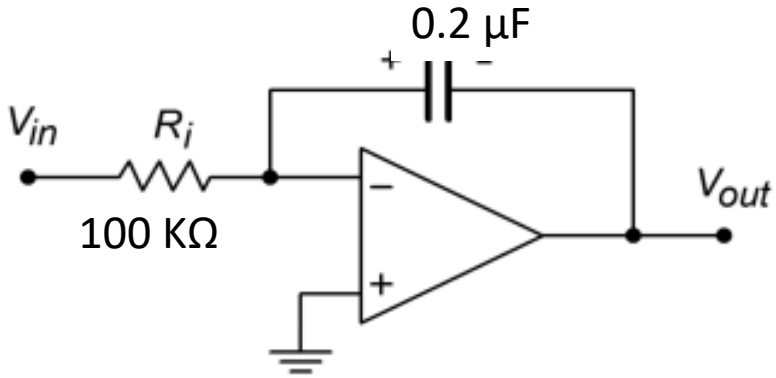
Solution:

$$V_{out}(t) = -\frac{1}{R_i C}\int V_{in}(t)\,dt$$

One possibility is as follows:

$$\frac{1}{RC} = 50$$

Let R = 100 kΩ, $C = \dfrac{1}{50\text{x}100\text{x}10^3} = 0.2\mu\text{F}$



Practice Problem

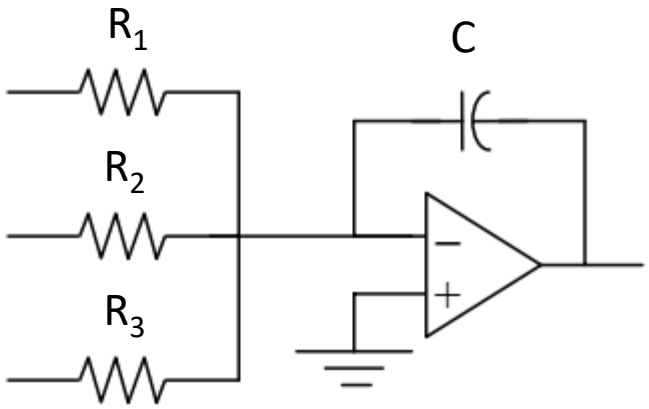
Show how you would use a single op amp to generate

$$v_0 = -\int (v_1 + 4v_2 + 10v_3)\,dt$$

If the integrating capacitor is $C = 2\ \mu$F, obtain other component values.

Solution:

By combining a summer with an integrator, we have the circuit below:



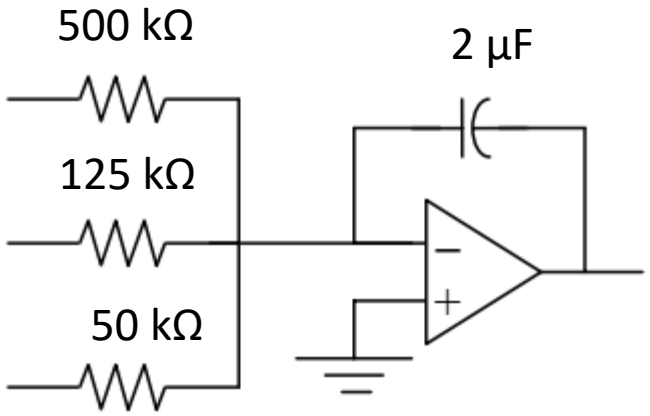
$$v_o = -\frac{1}{R_1 C}\int v_1 dt - \frac{1}{R_2 C}\int v_2 dt - \frac{1}{R_2 C}\int v_2 dt$$

For the given problem, C = **2μF**,

$R_1C = 1 \longrightarrow R_1 = 1/(C) = 10^6/(2) =$ **500 kΩ**

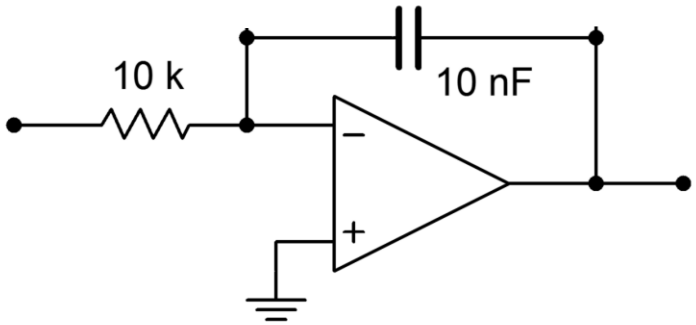
$R_2C = 1/(4) \longrightarrow R_2 = 1/(4C) = 500k\Omega/(4) =$ **125 kΩ**

$R_3C = 1/(10) \longrightarrow R_3 = 1/(10C) =$ **50 kΩ**



Practice Problem

Using the circuit give below, determine the output if the input is a 1 V peak sine wave at 5 kHz.



Solution:

The input signal as a function of time,

$$V_{in}(t) = 1 \sin 2\pi 5000 t$$

The output signal as a function of time,

$$V_{out}(t) = -\frac{1}{R_i C} \int V_{in}(t)\, dt$$

The output signal as a function of time,

$$V_{out}(t) = -\frac{1}{R_i C}\int V_{in}(t)\,dt$$

$$V_{out}(t) = -\frac{1}{10\,\mathrm{k}\times 10\,\mathrm{nF}}\int 1\sin 2\pi 5000\,t\,dt$$

$$V_{out}(t) = -10^4\int 1\sin 2\pi 5000\,t\,dt$$

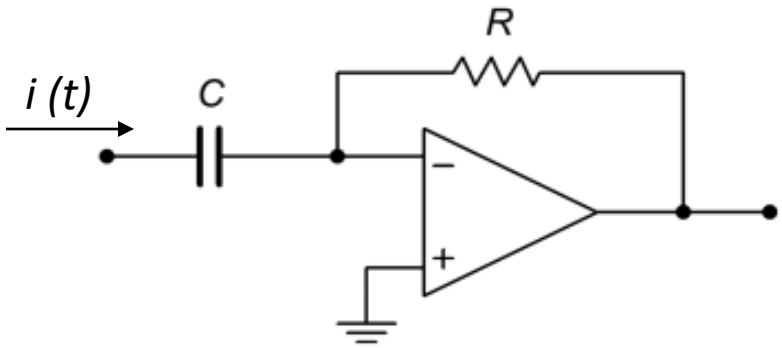
$$V_{out}(t) = -10^4\frac{1}{2\pi 5000}\int 2\pi 5000\sin 2\pi 5000\,t\,dt$$

$$V_{out}(t) = -0.318\left(-\cos 2\pi 5000\,t\right)$$

$$V_{out}(t) = 0.318\cos 2\pi 5000\,t$$

Differentiator

A **differentiator** is an op amp circuit whose output is proportional to the rate of change of the input signal.



$$i(t) = C\frac{dv(t)}{dt}$$

$$V_{out} = -V_{Rf} \qquad V_{Rf} = i\,R_f$$

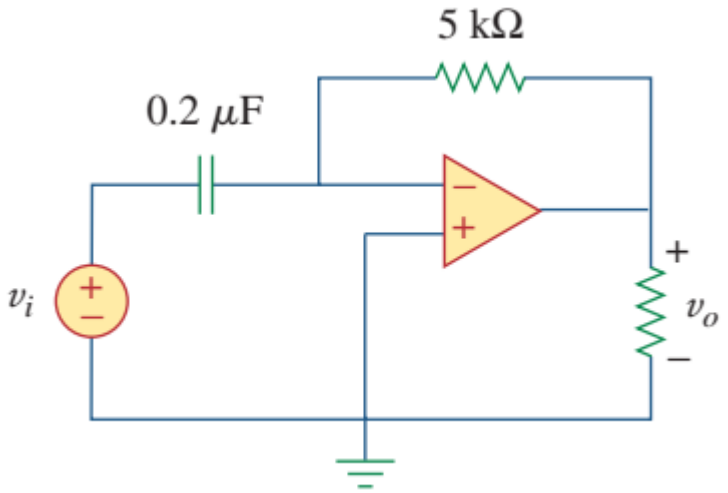
$$V_{out}(t) = -R_f C\frac{dv(t)}{dt}$$

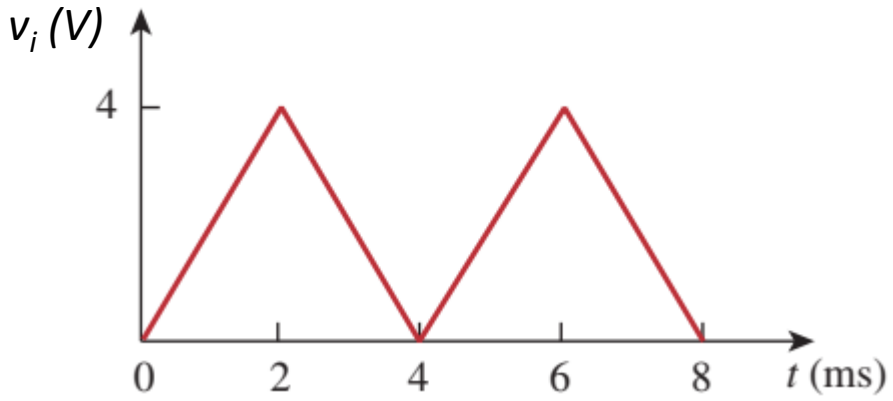
OR

$$v_o = -RC\frac{dv_i}{dt}$$

Practice Problem

Sketch the output voltage for the circuit, given the input voltage.
Take $v_0 = 0$ at $t = 0$.





Solution:

This is a differentiator with

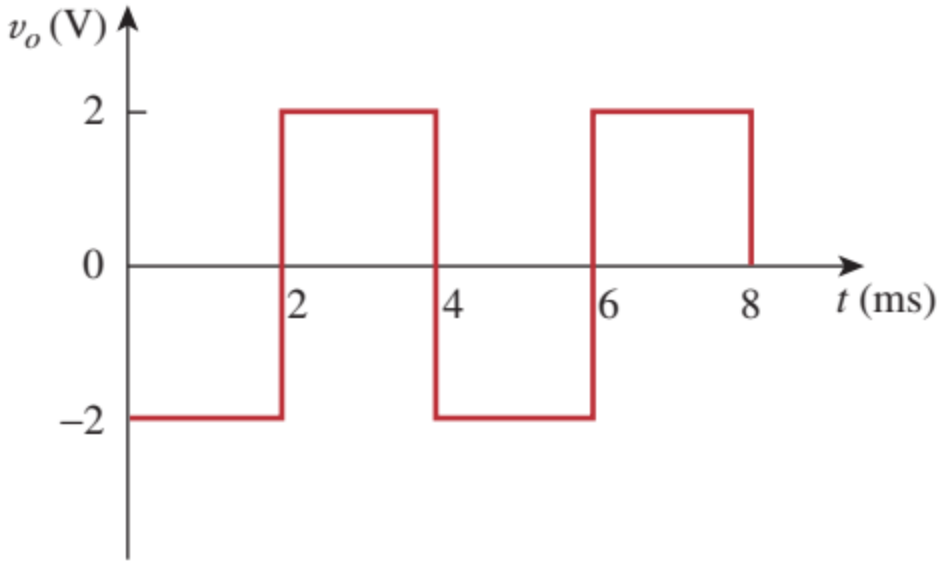
$$RC = 5 \times 10^3 \times 0.2 \times 10^{-6} = 10^{-3}\ \text{s}$$

For $0 < t < 4$ ms, we can express the input voltage

$$v_i = \begin{cases} 2000t & 0 < t < 2\ \text{ms} \\ 8 - 2000t & 2 < t < 4\ \text{ms} \end{cases}$$

This is repeated for $4 < t < 8$ ms.

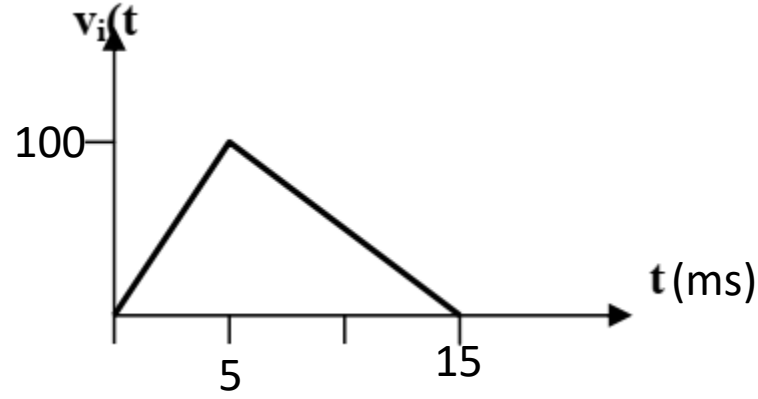
the output is $$v_o = -RC\frac{dv_i}{dt} = \begin{cases} -2\text{ V} & 0 < t < 2\text{ ms} \\ 2\text{ V} & 2 < t < 4\text{ ms} \end{cases}$$



Practice Problem

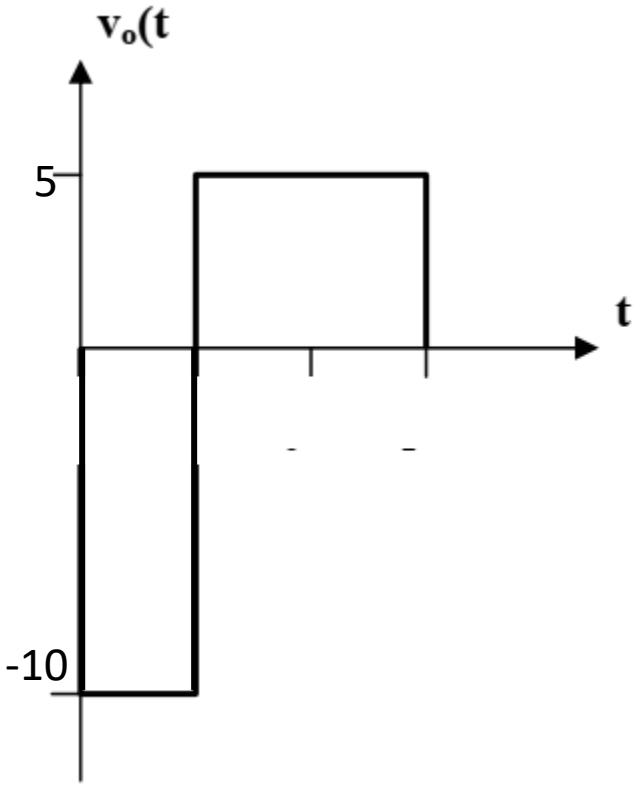
A voltage waveform has the following characteristics: a positive slope of 20 V/s for 5 ms followed by a negative slope of 10 V/s for 10 ms. If the waveform is applied to a differentiator with $\boldsymbol{R}$ = 50 kΩ, $\boldsymbol{C}$ = 10 μF, sketch the output voltage waveform.

Solution:



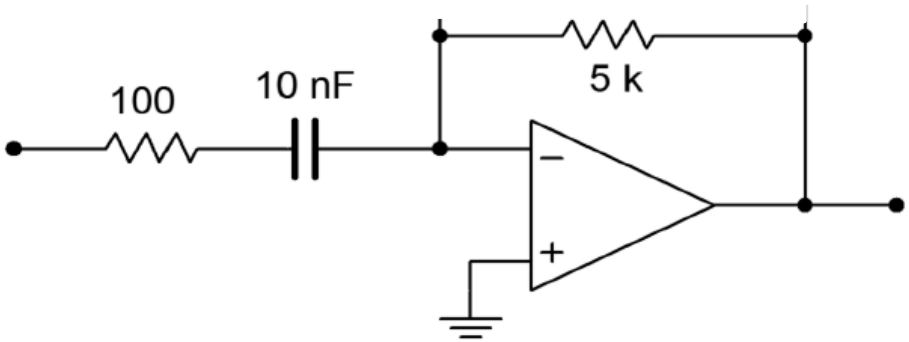
$$v_o = -RC\frac{dv_i}{dt},\quad RC = 50 \times 10^3 \times 10 \times 10^{-6} = 0.5$$

$$v_o = -0.5\frac{dv_i}{dt} = \begin{cases} -10, & 0 < t < 5 \\ 5, & 5 < t < 15 \end{cases}$$



Practice Problem

Sketch the output waveform for the circuit if the input is a 3 volt peak triangle wave at 4 kHz.



Solution:

The triangle wave may be broken into a positive-going portion and a negative-going portion. In either case, the total voltage change will be 6 V in one half-cycle. The period of the waveform is

$$T = \frac{1}{4\,\text{kHz}}$$

$$T = 250\,\mu\text{s}$$

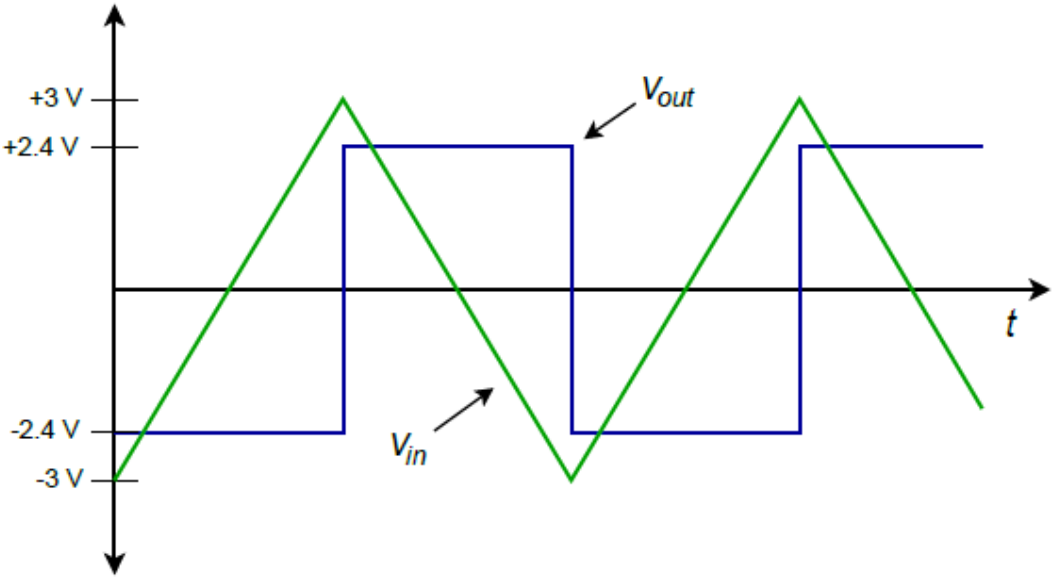
Therefore, for the positive going portion, a 6 volt change will be seen in 125 μs (−6 V in 125 μs for the negative going portion). The slope is

$$Slope = \frac{6\,\mathrm{V}}{125\,\mu\mathrm{s}}$$
$$Slope = 48000\,\mathrm{V/s}$$

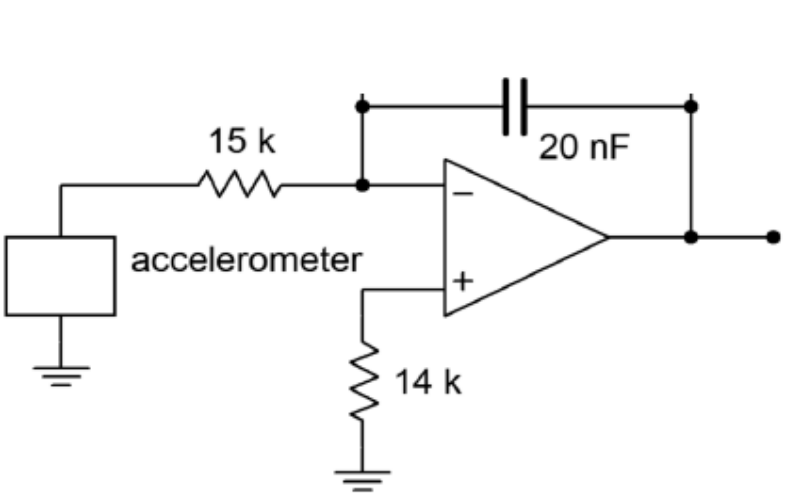
$$V_{in}(t) = 48000\,t$$

$$V_{out}(t) = -R_f C \frac{dV_{in}(t)}{dt}$$
$$V_{out}(t) = -5\,\mathrm{k} \times 10\,\mathrm{nF} \frac{d\,48000\,t}{dt}$$
$$V_{out}(t) = -2.4\,\mathrm{V}$$

During t = 0 through t = 125 µs, the output is −2.4 V. Differentiation of the second half of the wave is similar, but produces a positive output, +2.4 V. The result is a 4 kHz square wave that is 2.4 V peak. The resulting waveform is shown in Figure



Practice Problem



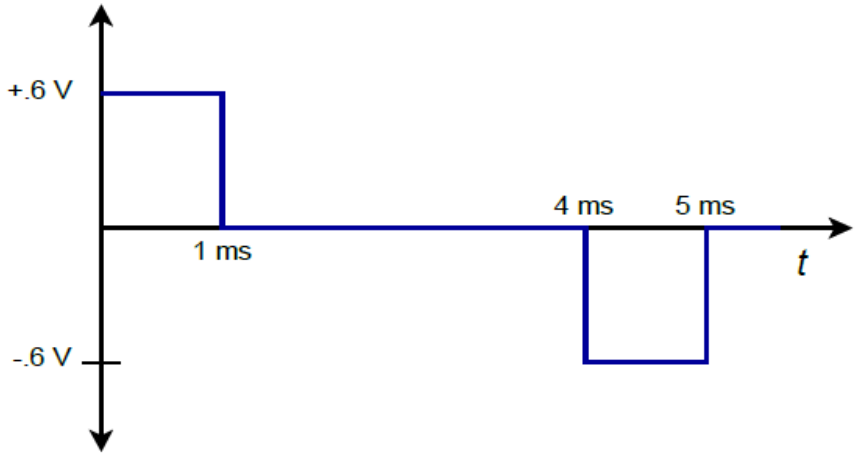


Figure shows an integrator connected to an accelerometer. If the accelerometer produces the voltage shown in Figure, determine the shape of the velocity curve.

Solution:

For the positive pulse (t = 0 to t =1 ms),

$$V_{out}(t) = -\frac{1}{R_i C}\int V_{in}(t)\,dt$$

$$V_{out}(t) = -\frac{1}{15\,\text{k}\times 20\,\text{nF}}\int_0^{10^{-3}} .6\,dt$$

$$V_{out}(t) = -3333\times 0.6\times t\Big|_{t=0}^{t=10^{-3}}$$

$$V_{out} = -2000\times 10^{-3}$$

$$V_{out} = -2\,\text{V}$$

For the time period between 1 ms and 4 ms, the input signal is zero, and thus the change in output potential will be zero.

For the period between 4 ms and 5 ms, a positive going ramp will be produced. Because only the polarity is changed relative to the first pulse, we can quickly find that the change is +2 V.

